

**ACTIVITY 9 — FACILITATOR GUIDE****Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen**

| TYPE          | DURATION | DMAIC PHASE |
|---------------|----------|-------------|
| Core Activity | 60 min   | IMPROVE     |

**Purpose of this activity**

Recommend improvement actions that address the proven root cause.

Generate and select countermeasures that attack the root cause you proved — not the symptom. Apply the core Lean improvement tools: 5S, poka-yoke, visual management, standard work and Kaizen.

**What learners produce**

A prioritised countermeasure set traced to proven root causes, a 5S plan, one poka-yoke design and draft standard work.

**Materials required**

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework

**Set-up**

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

**Scenario / briefing material**

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55

minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

## Files in this activity folder

### templates/ — the worksheets you complete:

- 5s-plan.csv (S, what\_it\_means\_here, action, owner)
- countermeasures.csv (root\_cause, countermeasure, type\_5S\_pokayoke\_standardwork, impact\_H\_M\_L, effort\_H\_M\_L, expected\_effect)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

### How to run it — step by step

| # | Instruction to learners                                                                          | Facilitator note                                                                         |
|---|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1 | Generate countermeasures for each proven root cause — quantity before quality.                   | Countermeasures that do not trace to a proven cause are solutions looking for a problem. |
| 2 | Prioritise using an impact vs effort matrix to find the quick wins.                              | Prevent beats detect — challenge any poka-yoke that is really just an alert.             |
| 3 | Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to digital work. | 5S applies to digital work too; do not accept 'this is an office, not a factory'.        |
| 4 | Design one poka-yoke that makes the error difficult or impossible to make.                       |                                                                                          |
| 5 | Draft standard work so the improved method is repeatable by anyone.                              |                                                                                          |
| 6 | Plan a Kaizen event or pilot to test the countermeasure at small scale first.                    |                                                                                          |
| 7 | Distinguish a containment countermeasure from a permanent solution.                              |                                                                                          |

### Suggested timing

| Minutes | Phase                                   |
|---------|-----------------------------------------|
| 15 min  | Generate countermeasures per root cause |
| 10 min  | Prioritise on impact vs effort          |
| 10 min  | Apply 5S thinking                       |
| 10 min  | Design one poka-yoke                    |
| 10 min  | Draft standard work                     |
| 5 min   | Plan the pilot                          |

**Total: 60 minutes**

### Discussion & decision prompts

1. Does your poka-yoke PREVENT the error or merely DETECT it? Which is stronger?
2. Which countermeasure is a containment, and which is permanent?
3. What would you pilot first, and how would you know it worked?

### Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the IMPROVE phase of DMAIC.
- Remind learners how this activity feeds the next phase of the improvement package.

Reflection questions:

- Where could a poka-yoke remove an error you personally make each week?
- What standard work exists in your team but is not written down?

### **Facilitator notes — what to watch for**

- Countermeasures that do not trace to a proven cause are solutions looking for a problem.
- Prevent beats detect — challenge any poka-yoke that is really just an alert.
- 5S applies to digital work too; do not accept 'this is an office, not a factory'.

### **Success criterion**

Every countermeasure traces back to a proven root cause, and your poka-yoke prevents rather than detects the error.

### **How this fits the course**

This activity builds the IMPROVE phase of DMAIC. Every activity reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000), so the outputs accumulate into one improvement package. Nothing is practised here that is not taught in the slides.